Name: ______________________________ **NetID:** ________________

ECON 1001 — Spring 2026 — Midterm 1

Version A

There are 21 multiple choice questions and 2 free response questions. Select the best answer using the models from the course. Each MC question and each FR subquestion is worth 4 points. There are a total of 112 possible points. ***All of your answers must go on the separate response sheet.***

0. Indicate your exam version, name, and NetID **on both sides** of your response sheet.

1. "Equity is more important than efficiency." This claim is

A) testable

B) causal

C) positive

D) normative

2. Brutus and Caesar have both consumed four cookies. Brutus's willingness to pay for a fifth cookie is \$5; Caesar's is \$3. This means

A) Brutus has higher income than Caesar.

B) Brutus and Caesar face different prices for buying cookies.

C) Brutus likes cookies more than Caesar.

D) None of the other options is *necessarily* true.

3. Bronson valued his watch at \$25. He sold it to Amirah for \$30, whose willingness to pay was \$50. This implies

A) This transaction was irrational.

B) This transaction increased total economic surplus by more than \$20.

C) The total increase in economic surplus from this transaction was \$20.

D) Amirah tricked Bronson.

4. Jeeves and Gertrude each own one bitcoin. The price of bitcoin has fallen. Jeeves sells his bitcoin; Gertrude does not sell hers. Which does **not** explain the difference in behavior (in our model)?

A) Jeeves is more sensitive to falls in price than Gertrude.

B) Gertrude is more optimistic about the future of bitcoin than Jeeves is.

C) Gertrude bought the bitcoin at a lower price than Jeeves did.

D) Gertrude is richer than Jeeves.

5. Consider the assumption of *diminishing marginal benefit* from consuming cookies. It implies:

A) It does not necessarily imply any of the other options.

B) You *always* benefit from an additional cookie, but each additional cookie adds less benefit than the previous one.

C) Eventually an additional cookie makes you worse off.

D) If cookies were free, you would never stop consuming them.

6. Bobby is definitely buying a \$400,000 house. He has \$400,000 in savings available today. He can either (i) pay \$400,000 today from his savings, or (ii) take out a 15-year loan that would require him to make 15 equal annual payments, and the sum of those payments would be \$500,000. Ignoring taxes and risk, Bobby should:
   A) Take the loan and pay \$500,000 over 15 years.
   B) It depends on how much Bobby likes the house.
   C) Pay \$400,000 today and not take the loan.
   D) It depends on other available uses of Bobby's savings (i.e., what return he could earn elsewhere).

7. In June 2024, the price of a Labubu doll was \$10 and 10,000 were sold. In June 2025, the price was \$20 and 100,000 were sold. Which is the best explanation?
   A) The demand curve shifted right.
   B) The demand curve slopes upward.
   C) This is a violation of the law of demand.
   D) Economic theory does not apply to this case.

8. For Brillo, ramen is an inferior good and sushi is a normal good. When Brillo's income rises, holding prices fixed, what happens to his *total spending on ramen and sushi*?
   A) It goes up.
   B) It goes down.
   C) It stays the same.
   D) Can't say.

9. A strike raised the price of flights to New York. The price of bus tickets to New York also rose. *Ceteris paribus*, this suggests:
   A) The cross-price elasticity of demand for bus tickets with respect to flight prices is positive.
   B) Demand for bus tickets to New York shifted right for reasons unrelated to flights.
   C) Flights and buses to New York are inferior goods.
   D) Flights and buses to New York are complements.

10. Bread and cookies use many of the same inputs. When the price of cookies rises, Raphael's Bakery produces fewer loaves of bread. This is best described as:
   A) Bread and cookies are complements in production.
   B) Bread and cookies are substitutes for consumers.
   C) Raphael's supply curve for bread shifts left.
   D) A mistake—cookie prices should not affect Raphael's supply of bread.

11. Suppose Taylor Swift concerts always sell out because the ticket price is below the equilibrium price, and tickets cannot be resold. The supply curve of tickets is perfectly inelastic (vertical). If Taylor raises the ticket price to the equilibrium level, it would be:
   A) More efficient, because the fans with the highest willingness to pay would be the ones to get the tickets.
   B) Less efficient, because fewer of her most devoted fans would attend her shows.
   C) More efficient, because Taylor would make more money.
   D) Less efficient, because on average richer people would attend her shows.

12. Suppose the demand for snowplowing services increases and the supply also increases. Then it is possible that in the new equilibrium:
A) ■ The price of snowplowing services increases.
B) ⋄ The price of snowplowing services does not change.
C) The quantity of snowplowing services does not change.
D) Both ⋄ and ■ are possibilities.

13. Each week, Buster buys two chocolate bars for $5 each and two bags of coffee beans for $10 each. A snowstorm caused the prices of both goods to increase by $3. As a result, Buster reduced his purchases to one chocolate bar and one bag of coffee beans. Assuming we can treat each market separately (*ceteris paribus*), this implies:
A) We can't compare the two elasticities because it depends on substitutes for those goods.
B) Buster's price elasticities of demand for the two goods are the same.
C) Buster's demand for chocolate is more elastic than his demand for coffee beans.
D) Buster's demand for chocolate is less elastic than his demand for coffee beans.

14. Bus tickets are an inferior good for Dianna. When her income increases, her demand curve for bus tickets shifts left. Suppose the new demand curve is parallel to the original (same slope). At the original bus-fare price (unchanged), what happens to Dianna's price elasticity of demand for bus tickets?
A) It becomes more elastic.
B) It becomes less elastic.
C) It stays the same.
D) Cannot be determined without knowing the price of bus tickets.

15. Consider the perfectly competitive market for widgets, which buyers pay for exclusively with credit cards. Suppose credit card companies start charging buyers an extra $1 for each widget they purchase. Assuming that neither supply nor demand is perfectly elastic or inelastic, in the new equilibrium, the price buyers have to pay is:
A) More than $1 higher than the original price.
B) Can't say without more information about the suppliers' prices.
C) Exactly $1 higher than the original price.
D) Higher than the original price, but not $1 higher.

16. Saachi sells trinkets, which cost her $2 each to make (i.e., $MC = 2$). Her economist tells her that demand for her product is a straight-line, downward-sloping demand curve and that, at the current price Saachi has set, demand is elastic (with respect to price). To increase her *profit* (not just total revenue), Saachi should:
A) Can't say without more information.
B) Increase her price so she sells fewer goods at a higher price.
C) Decrease her price so she sells more goods at a lower price.
D) Keep her price the same.

17. A mayor had to choose between two policies: Policy A would have increased surplus for the rich by \$150 and *decreased* surplus for the poor by \$20. Policy B would have increased surplus for the rich by \$80 and *increased* surplus for the poor by \$20. Which statement is correct?

A) Either policy A or policy B could be more efficient, depending on the mayor's values.

B) If the mayor is primarily concerned with equity, then she should choose policy B even if she can redistribute economic surplus without reducing the total amount.

C) None of the other options is correct.

D) If, after the policy is chosen, the mayor can redistribute economic surplus however she likes (without reducing total surplus), then she should choose policy A.

18. Some time ago, Trisha gave up one shift at work (worth \$40) to wait in line to buy a \$50 ticket to see a Bad Bunny concert. On the night of the concert, someone offered her \$100 for the ticket, but she declined (recall our class convention: if she valued it at exactly \$100 she would have sold). Then someone offered her \$200, and she sold the ticket. We know that Trisha:

A) has a willingness to pay of ≥ \$110 to see the Bad Bunny concert.

B) ♡ and ⋄.

C) ♡ has a willingness to pay of ≥ \$190 to see the Bad Bunny concert.

D) ⋄ has a willingness to pay greater than \$100 but less than or equal to \$200 (i.e., $\$100 < WTP \leq \$200$) to see the Bad Bunny concert.

19. The government of Theastan wants to raise revenue using a 5 percent tax on widgets. If it wants to minimize the impact of the tax on **consumer surplus**, it should:

A) put the full 5 percent tax on sellers.

B) Whatever the government does, the impact on buyers will be the same.

C) put the full 5 percent tax on buyers.

D) put a 2.5 percent tax on buyers and a 2.5 percent tax on sellers.

20. See Figure 1 on page 6. Suppose the government imposes a price ceiling of 7. What is the quantity exchanged in the market?

A) Can't say.

B) 25

C) 20

D) 15

21. See Figure 1 on page 6. Suppose a quota restricts the market to at most 15 units sold. The price adjusts accordingly. Which area represents the *change* in producer surplus due to the quota (PS after − PS before)?

A) $B - G$

B) $B + F + K$

C) $-B - C$

D) $C + G$

Free Response

Answer the problems on the back of your bubble sheet.

22. Consider the market for widgets, with demand and supply given by:

$$Q_D = 24 - 2P \qquad Q_S = 12 + 2P$$

Answer the following:

a) Sketch the supply and demand diagram. Be sure to label the *price intercepts* (i.e., the intercepts on the vertical axis, where $Q = 0$) of both curves.

b) Solve for the equilibrium price, P^*. Show your work.

c) Suppose the government imposes a per-unit tax of 2 on sellers. What is the new quantity with the tax, Q_t?

d) With the tax in place, what price do buyers pay, P_b?

e) Does the tax burden fall more on buyers or sellers, or is it split evenly? (Compare the change in P_b to the change in the price sellers keep, P_s.) Briefly explain.

23. Two firms, A and B, produce the same good. Their marginal costs are shown in the table below.

Q	MC_A	MC_B
0	0	0
1	1	2
2	2	4
3	3	6
4	4	8
5	5	10

a) First suppose the market price is $P = 4$. How many units does each firm supply? (Assume whole-number quantities. As usual, each firm produces when $MC \le P$ and stops when the next unit would have $MC > P$.)

b) Instead suppose the total quantity to be produced is fixed at $Q_A + Q_B = 6$. If production is efficient (total cost minimized), what are Q_A and Q_B? Assume quantities must be whole numbers. Explain briefly.

Do not turn over this exam until instructed to.

Write your name and NetID on **both sides** of your response sheet.

You may write on this exam packet, however, only your response sheet will be graded. For the free response problems, you must show your work to receive full credit.

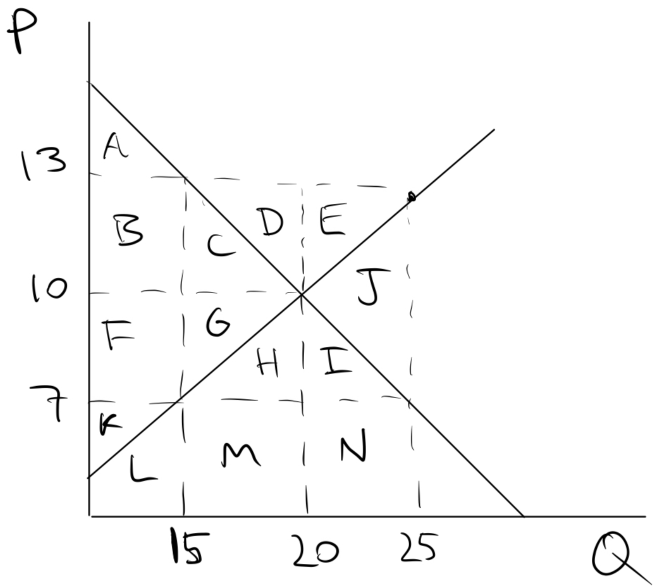

Figure 1: Supply and Demand